

# Binomial Theorem, Inclusion/Exclusion

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# Overview

The Binomial Theorem (15.7)

Inclusion-Exclusion (15.12)

- Union of Two Sets (15.12.1)

- Union of Three Sets (15.12.2)

- Sequences with 42, 04, or 60 (15.12.3)

- Union of  $n$  Sets (15.12.4)

## Historical perspective

*I'm very well acquainted too with matters mathematical,  
I understand equations, both the simple and quadratical,  
About binomial theorem I am teeming with a lot o' news—  
With many cheerful facts about the square of the hypotenuse.*

From “Modern Major General” by Arthur Sullivan and William Schwenck Gilbert (1879)

## Binomials to powers: Examples

$$\begin{aligned}
 (a + b)^2 &= aa + ab + ba + bb \\
 &= a^2 + 2ab + b^2
 \end{aligned}$$

$$\begin{aligned}
 (a + b)^3 &= aaa + aab + aba + abb \\
 &\quad + baa + bab + bba + bbb \\
 &= a^3 + 3a^2b + 3ab^2 + b^3
 \end{aligned}$$

$$\begin{aligned}
 (a + b)^4 &= aaaa + aaab + aaba + aabb \\
 &\quad + abaa + abab + abba + abbb \\
 &\quad + baaa + baab + baba + babb \\
 &\quad + bbaa + bbab + bbba + bbbb \\
 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4
 \end{aligned}$$

How about  $(a + b)^n$ ? How many terms consist of exactly  $k$   $bs$ ?

Since it's all combinations of an  $a$  and  $b$  in each position, there are  $\binom{n}{k}$  such terms.

# Binomial theorem

**Theorem:** For all  $n \in \mathbb{N}$ ,  $a, b \in \mathbb{R}$ ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

Sometimes  $\binom{n}{k}$  called the *binomial coefficient* because of this connection.

# Pets and sets

$S$ : Set of all students in CS0220.

$D \subseteq S$ : Set of all students in CS0220 who have a pet dog.

$C \subseteq S$ : Set of all students in CS0220 who have a pet cat.

$D \cup C$ : Set of all students in CS0220 who have a pet dog *or* cat.

$|D \cup C| = |D| + |C|$ ? Handles people who have neither correctly.  
Handles people who have one kind of pet correctly. Messes up on  
people who have both.

# Formulas for union

What's wrong with each formula for  $|C \cup D|$ ?

- ▶  $|C| + |D|$ ? Double counted people who have both.
- ▶  $|C - D| + |D - C|$ ? Skipped people who have both.
- ▶  $|C - D| + |D - C| + |C \cap D|$ ? Actually, that should work.  
But, set difference can be tricky.
- ▶  $|C| + |D| - |C \cap D|$ ? Nailed it. Correct for double counting

## Inclusion-Exclusion rule for two sets

**Rule:** For two sets  $S_1$  and  $S_2$ ,

$$|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2|.$$

Example:

- ▶  $S_1 = \{ \text{Morley, Kyran} \}$ : HTAs with a *y* in their name.
- ▶  $S_2 = \{ \text{Morley, Will} \}$ : HTAs with an *l* in their name.
- ▶  $S_1 \cap S_2 = \{ \text{Morley} \}$ : HTAs with both a *y* and an *l* in their name.
- ▶  $S_1 \cup S_2 = \{ \text{Morley, Kyran, Will} \}$ : HTAs with either a *y* or an *l* in their name.
- ▶  $|\{ \text{Morley, Kyran, Will} \}| = |\{ \text{Morley, Kyran} \}| + |\{ \text{Morley, Will} \}| - |\{ \text{Morley} \}|$



## Generalize to three sets

$S$ : Set of all students in CS0220.

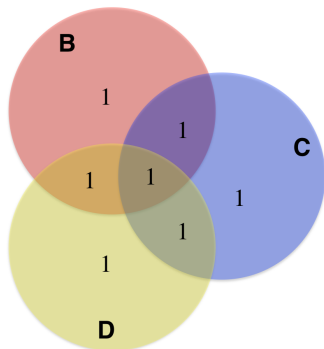
$D \subseteq S$ : Set of all students in CS0220 who have a pet dog.

$C \subseteq S$ : Set of all students in CS0220 who have a pet cat.

$B \subseteq S$ : Set of all students in CS0220 who have a pet bunny.

How express  $|B \cup C \cup D|$  in terms of size of *intersections* of sets?

# Visual analysis



$$|B \cup C \cup D| = |B| + |C| + |D| - |B \cap C| - |B \cap D| - |C \cap D| + |B \cap C \cap D|$$

# Inclusion-Exclusion rule for three sets

**Rule:** For three sets  $S_1, S_2, S_3$ ,

$$|S_1 \cup S_2 \cup S_3| = |S_1| + |S_2| + |S_3| \\ - |S_1 \cap S_2| - |S_1 \cap S_3| - |S_2 \cap S_3| \\ + |S_1 \cap S_2 \cap S_3|.$$

Example:

- ▶  $S_1 = \{ \text{Morley, Kyran} \}$ : HTAs with a  $y$  in their name.
- ▶  $S_2 = \{ \text{Morley, Will} \}$ : HTAs with an  $l$  in their name.
- ▶  $S_3 = \{ \text{Morley, Kyran} \}$ : HTAs with an  $r$  in their name.
- ▶  $S_1 \cap S_2 \cap S_3 = \{ \text{Morley} \}$ : HTAs with a  $y$  and an  $l$  and an  $r$  in their name.
- ▶  $|\{ \text{Morley, Kyran, Will} \}| = |\{ \text{Morley, Kyran} \}| + |\{ \text{Morley, Will} \}| + |\{ \text{Morley, Kyran} \}| - |\{ \text{Morley} \}| - |\{ \text{Morley, Will} \}| - |\{ \text{Morley, Kyran} \}| + |\{ \text{Morley} \}|$

## Sets of permutations

In how many permutations of the set  $\{0, 1, 2, \dots, 9\}$  do either 4 and 2, 0 and 4, or 6 and 0 appear consecutively?

Which of these permutations has this property?

- ▶ (4, 6, 5, 0, 1, 8, 3, 2, 9, 7) nope.
- ▶ (0, 4, 6, 1, 8, 5, 9, 3, 7, 2) 04!
- ▶ (3, 4, 2, 0, 5, 6, 1, 9, 8, 7) 42!
- ▶ (3, 9, 4, 1, 2, 7, 0, 5, 6, 8) nope.
- ▶ (0, 2, 6, 3, 7, 8, 4, 9, 5, 1) nope.

$$2/5 \times 10! = 1451520.$$

$P_{60}$ : permutations of 0 through 9 that contain 60.

$P_{04}$ : permutations of 0 through 9 that contain 04.

$P_{42}$ : permutations of 0 through 9 that contain 42.

Want:  $|P_{60} \cup P_{04} \cup P_{42}|$ .

# Inclusion-exclusion, constrained permutation

$$\begin{aligned} & |P_{60} \cup P_{04} \cup P_{42}| \\ &= |P_{60}| + |P_{04}| + |P_{42}| \\ &\quad - |P_{60} \cap P_{04}| - |P_{42} \cap P_{04}| - |P_{60} \cap P_{42}| \\ &\quad + |P_{60} \cap P_{04} \cap P_{42}| \end{aligned}$$

$$|P_{60}| = ?$$

Clever trick: In  $P_{60}$ , can view “60” as a unit. So, each element of  $P_{60}$  is a permutation of  $\{1, 2, 3, 4, 5, 7, 8, 9, 60\}$ . Therefore,  $|P_{60}| = 9!$ .  $|P_{04}| = 9!$ .  $|P_{42}| = 9!$ .

## Pairwise intersections

$$\begin{aligned}
 & |P_{60} \cup P_{04} \cup P_{42}| \\
 &= |P_{60}| + |P_{04}| + |P_{42}| \\
 &\quad - |P_{60} \cap P_{04}| - |P_{42} \cap P_{04}| - |P_{60} \cap P_{42}| \\
 &\quad + |P_{60} \cap P_{04} \cap P_{42}| \\
 &= 3 \times 9! \\
 &\quad - |P_{60} \cap P_{04}| - |P_{42} \cap P_{04}| - |P_{60} \cap P_{42}| \\
 &\quad + |P_{60} \cap P_{04} \cap P_{42}|
 \end{aligned}$$

$|P_{60} \cap P_{04}| = ?$  Trick works again! Can view “604” as a unit. So, each element is a permutation of  $\{1, 2, 3, 5, 7, 8, 9, 604\}$ .

Therefore,  $8!$ .

$|P_{42} \cap P_{04}| = ?$  Trick works again! Can view “042” as a unit. So,  $8!$ .

$|P_{60} \cap P_{42}| = ?$  Trick fails! Wait, no, just changes. Now, each element is a permutation of  $\{1, 3, 5, 7, 8, 9, 60, 42\}$ . Still  $8!$ .

## Three-way intersection

$$\begin{aligned}
 & |P_{60} \cup P_{04} \cup P_{42}| \\
 &= |P_{60}| + |P_{04}| + |P_{42}| \\
 &\quad - |P_{60} \cap P_{04}| - |P_{42} \cap P_{04}| - |P_{60} \cap P_{42}| \\
 &\quad + |P_{60} \cap P_{04} \cap P_{42}| \\
 &= 3 \times 9! - 3 \times 8! \\
 &\quad + |P_{60} \cap P_{04} \cap P_{42}|
 \end{aligned}$$

$|P_{60} \cap P_{04} \cap P_{42}| = ?$ . Yay, trick works again! Can view “6042” as a unit. So, each element is a permutation of  $\{1, 3, 5, 7, 8, 9, 6042\}$ . Therefore,  $7!$ .

$$|P_{60} \cup P_{04} \cup P_{42}| = 3 \times 9! - 3 \times 8! + 7! = 972720.$$

Sampling estimate was actually pretty good. Guessed 20% of permutations, but it's closer to 27%.

# n-way Inclusion-Exclusion

$$|S_1 \cup S_2 \cup \cdots \cup S_n| =$$

the sum of the sizes of the individual sets  
 minus the sizes of all two-way intersections  
 plus the sizes of all three-way intersections  
 minus the sizes of all four-way intersections  
 plus the sizes of all five-way intersections, etc.

Hyper-mathy version:

$$\left| \bigcup_{i=1}^n S_i \right| = \sum_{X \in \mathcal{P}([1,n]) - \emptyset} (-1)^{|X|+1} \left| \bigcap_{i \in X} S_i \right|$$